Digital Literacy as a Key Factor in Marketing Education and Business-Academy Relationship: A Digital Transformation Approach

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Abstract. This study analyses the role of digital literacy in marketing and business science curricula, focusing on its impact on students' preparation for the digital labour market. It aims to understand the challenges and opportunities in acquiring digital skills and collaboration between the academy and business in the context of digital transformation. The research is based on a qualitative analysis of focus groups with marketing graduate students, exploring their perceptions about the adequacy of academic curricula in the face of digital market demands. The results reveal a significant gap between academic training and the companies' practical needs and show the importance of continuous and closer collaboration between the educational and business sectors. Despite efforts to integrate digital skills, students feel insufficiently prepared to face the challenges of the digital market. The study presents practical recommendations to adjust marketing curricula, proposing a constant update of digital tools and a greater integration of practical projects and business collaborations to improve students' preparation and facilitate the process of digital transformation in companies while also contributing to the theoretical understanding of digital literacy as a pivotal element in fostering collaboration between academia and the business sector during digital transformation processes.

Keywords: Digital Literacy · Digital Skills · Digital Transformation · Marketing Education

1 Introduction

Digital transformation is redefining organisational operations and professional skills, driven by the integration of technologies such as Artificial Intelligence (AI), the Internet of Things (IoT), and Big Data, increasing the need for digital literacy in marketing and business sciences (Abdullah et al., 2024; Calderon-Monge & Ribeiro-Soriano, 2023). Academia plays a fundamental role in preparing students by ensuring curricula are

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aligned with market demands and integrating digital competencies to support organisational transformation (Blayone, 2018; Khan et al., 2022; Radzikhovska, 2021). However, challenges arise when academic programs do not reflect the digital skills required by businesses, creating gaps in adaptation to the technological environment (Petkovics, 2018; Radzikhovska, 202; Rêgo et al., 2024).

Collaboration between academia and the business sector is essential for innovation and competitiveness; however, some courses use outdated tools and provide limited experiential learning opportunities, hindering students' transition into the digital market (He et al., 2024; Khan et al., 2022; Petkovics, 2018). This study examines the integration of digital literacy in marketing education and its contribution to professional preparedness and digital transformation in organisations (Bezpartochnyi, 2023; Szopik-Depczyńska et al., 2023; Viana & Gomes, 2025).

The research is structured around three propositions: (1) integrating digital literacy into curricula enhances employability and aligns skills with market needs; (2) strengthening academia-business collaboration facilitates curricular alignment with industry requirements; (3) incorporating digital skills into marketing education reinforces strategic and technical capabilities, fostering innovation and business adaptability in digital markets.

Based on these propositions, the study is guided by four specific research objectives that operationalise the investigation and ensure alignment between theoretical assumptions and empirical exploration: (1) identify the challenges and opportunities related to the acquisition of digital skills and academic-business collaboration in the context of digital transformation; (2) Explore student perceptions of curriculum adequacy in response to digital market demands; (3) provide practical recommendations for curriculum adjustment based on industry expectations; and (4) contribute theoretically to the understanding of digital literacy as a catalyst for collaboration between academia and companies.

2 Theoretical Background

2.1 Digital Literacy and Digital Transformation in Business and Marketing Education

Digital literacy is fundamental to digital transformation in education and business, aligning technical proficiency with industry needs (Farias-Gaytan et al., 2023; Khan et al., 2022; Verhoef et al., 2021). As businesses accelerate digitalisation, demand for AI, IoT, and Big Data expertise is reshaping curricula (Abdullah et al., 2024; Akinwalere & Ivanov, 2022; Njeru, 2023). Academy must integrate these competencies to prepare students for evolving labour market expectations (Langan et al., 2019; Rêgo et al., 2024; van Laar et al., 2020; Viana & Gomes, 2025). Digital transformation enhances organisational efficiency and competitiveness, reinforcing the strategic role of digital literacy in business sustainability (Anayat & Rasool, 2024; Calderon-Monge & Ribeiro-Soriano, 2023).

Embedding digital competencies into curricula bridges the gap between academic knowledge and business applications, improving graduates' employability and ability to contribute to digital transformation (Petkovics, 2018; Radzikhovska, 2021). Theoretical

instruction alone is insufficient, and practical experience with digital tools is used for aligning skills with industry requirements (Abdullah et al., 2024; Ye et al., 2024). Collaboration between academia and business ensures that curricula incorporate relevant technologies and real-world applications (Abdullah et al., 2024; Petkovics, 2018). The relationship between these partnerships can support graduates' workforce transition and foster more effective digital transformation strategies (Radzikhovska, 2021; Rêgo et al., 2024).

2.2 Challenges of Academia-Companies Collaboration in the Digital Era

Academia-business collaboration drives innovation by aligning education with industry expectations (Abdullah et al., 2024; Anayat & Rasool, 2024). However, significant gaps persist, particularly in applied digital skills (He et al., 2024; Petkovics, 2018; Maheshwari & Naik, 2024). Structured partnerships enhance students' preparedness for digital roles and contribute to corporate innovation (Calderon-Monge & Ribeiro-Soriano, 2023; Supriadi et al., 2024).

The Triple Helix Theory identifies universities, businesses, and governments as key actors in technological advancement, with higher education playing a central role in digital transformation (Langan et al., 2019; Supriadi et al., 2024). However, curriculum misalignment with industry expectations remains a critical barrier, particularly regarding practical digital competencies (He et al., 2024; Rêgo et al., 2024). Employers frequently highlight graduates' lack of essential digital skills, limiting organisations capacity to adapt to technological change (Radzikhovska, 2021; van Laar et al., 2020).

Academy must proactively revise curricula to incorporate emerging digital trends while fostering stronger industry engagement (Bezpartochnyi, 2023). Ensuring students develop digital expertise through experiential learning strengthens their ability to support digital transformation initiatives (Khan et al., 2022; Radzikhovska, 2021).

2.3 Digital Skills and Marketing-Oriented Strategies for Digital Transformation

Organisations leveraging digital tools can better anticipate consumer behaviour and market shifts (Ersoy & Tehci, 2023; Radzikhovska, 2021). Beyond adopting new technologies, data-driven marketing enhances decision-making and customer engagement (Anayat & Rasool, 2024; Ye et al., 2024). In this way, the academy must integrate data analytics, automation tools, and strategic marketing applications to ensure alignment with business needs because, as digital transformation advances, organisations require professionals capable of applying digital solutions to optimise business operations and customer experiences (Farias-Gaytan et al., 2023; He et al., 2024; Rêgo et al., 2024; Verhoef et al., 2021).

Marketing innovation increasingly relies on real-time consumer insights to refine strategies and maintain competitive advantage (Abdullah et al., 2024; Szopik-Depczyńska et al., 2023). Companies must continuously develop technological capabilities and strategic frameworks that support long-term digital adaptation (Gurjar et al., 2024; Ye et al., 2024; van Laar et al., 2020). Adopting an innovation-driven culture within marketing teams is essential for integrating digital tools into core business functions (Njeru, 2023; Szopik-Depczyńska et al., 2023).

3 Methodology

3.1 Empirical Context

The theoretical analysis reinforces the need for collaboration between academia and companies to face the challenges of digital transformation.

To explore this issue, the research was conducted in a Portuguese higher education institution, focusing on marketing students of the last year of their degree. The total population consisted of 45 eligible students. The 12 selected participants are considered acceptable in intentional sampling for qualitative research to obtain information-rich cases (Palinkas et al., 2015). The participants' experience in marketing and participation in projects related to the area also ensured the relevance of the perspectives.

The research design is structured around three core propositions, each aligned with a guiding research question. Table 1 presents this alignment, demonstrating how the interview script was designed to explore and potentially confirm each proposition through students' perceptions and experiences.

The focus group questions were developed to explore and support the propositions, and each item of the semi-structured script corresponds to one or more research propositions to ensure conceptual coherence between data collection and the theoretical framework.

3.2 Data Collection

Qualitative Approach and Focus Groups. A qualitative approach was adopted to explore students' perceptions of digital literacy and its role in preparing them for collaboration with businesses in digital transformation. This method provides in-depth insights into the complexities of digital skills integration and academia-business collaboration (Creswell & Poth, 2016; Denzin & Lincoln, 1996, 2011). Focus groups were the primary data collection method, capturing collective perspectives on digital literacy (Bryman, 2016; Krueger, 2014). Widely applied in research, focus groups help explore experiences and opinions, making them suitable for examining students' views on digital competencies and their relevance in business settings (Beck et al., 1986; Hair et al., 2017; Wilkinson, 1998a, 1998b, 2014). Online focus groups via Zoom (June 2024) offered flexibility and cost-effectiveness, enabling accurate data collection (Richard et al., 2021).

Sample and Participant Selection. The study involved twelve (12) final-year marketing students selected based on exposure to digital marketing and practical experiences such as internships or academy-business projects. Purposive sampling ensured that participants had relevant insights into digital tools and business collaboration (Creswell et al., 2009; Palinkas et al., 2015). Preparing to enter the job market, final-year students are well-positioned to evaluate how their education aligns with industry expectations (Verhoef et al., 2021). Participants, aged 20–23, were primarily full-time students, with one employed part-time in a communication agency. All had marketing training, but their exposure to digital tools varied, with some having more opportunities to use advanced technologies. A second focus group with six (6) participants was planned, but one was absent. Wilkinson (1998a, 1998b, 2014), and Morrow (2005), recommended that its

Table 1. Research propositions and questions.

Research Questions (RQ)	Authors	Description	Research Proposition (RP)
RQ1: How does integrating digital literacy into marketing education impact students' preparation for digital transformation?	Abdullah et al. (2024), Khan et al. (2022), Radzikhovska (2021), Rêgo et al. (2024)	The integration of digital literacy into academic curricula improves students' ability to respond to the demands of digital transformation in organisations, promoting their employability and aligning their skills with real market needs.	RP1
RQ2: What role does academia-business collaboration play in aligning academic curricula with digital market demands?	Calderon-Monge & Ribeiro-Soriano (2023), Farias-Gaytan et al. (2023), He et al. (2024), Rêgo et al. (2024), Supriadi et al. (2024)	Effective collaboration between academia and business promotes the alignment of academic curricula with market demands, leading to the development of practical digital skills and supporting the successful implementation of digital transformation initiatives	RP2
RQ3: How can digital skills in marketing education contribute to organisational competitiveness in the digital era?	Abdullah et al. (2024), Akinwalere & Ivanov (2022), Radzikhovska (2021), Szopik-Depczyńska et al. (2023), Ye et al. (2024)	Incorporating digital skills into marketing education gives students the strategic and technical skills to leverage advanced technologies, promoting innovation and improving organisations' ability to adapt to digital markets.	RP3

results were excluded from the final discussion to avoid overgeneralisation despite aligning with prior findings. The second focus group was initially designed to validate and

complement the findings from the first group. Due to incomplete participation, its exclusion did not affect the fulfilment of research objectives, as thematic saturation had already been reached and confirmed through triangulation with initial codes and categories.

3.3 Focus Group Structure and Facilitation

The moderator followed a semi-structured guide to ensure consistency across groups while allowing flexibility for participants to elaborate on their experiences (Kallio et al., 2016; Palermo & Hernandez, 2020). Ethical considerations were addressed, with participants informed of recording procedures and consent obtained before sessions began (Richard et al., 2021). Zoom provided a structured environment that facilitated discussion and remote participation while offering transcription tools for analysis. The online format was beneficial in ensuring diverse contributions and maintaining engagement despite geographical constraints (Morrow, 2005).

Table 2 maps the semi-structured script used during the focus groups to the theoretical constructs and authors, ensuring conceptual coherence between data collection and analytical categories.

Table 2. Semi-structured focus group script.

Constructs	Items	Authors
Perceptions of digital literacy and collaboration	How do you define digital literacy in business and marketing?	Khan et al. (2022); Rêgo et al. (2024)
	Importance of digital literacy in digital transformation processes.	Calderon-Monge & Ribeiro-Soriano (2023); Maheshwari & Naik (2024)
Experiences with digital technology	Digital technology experiences during coursework and business projects.	Blayone (2018); Farias-Gaytan et al. (2023)
	Influence of technology on learning and job market readiness.	Akinwalere & Ivanov (2022); Radzikhovska (2021)
Digital competencies and academia-business collaboration	Key digital competencies for collaboration in digital transformation.	Abdullah et al. (2024); Rêgo et al. (2024)
	Do students feel prepared to collaborate with companies after graduation?	Calderon-Monge & Ribeiro-Soriano (2023); Supriadi et al. (2024)
Digital transformation in curriculum	How should marketing curricula evolve to prepare students for digital transformation?	Khan et al. (2022); Petkovics (2018)

(*continued*)

Table 2. (*continued*)

Constructs	Items	Authors
	Alignment of the curriculum with digital labour market needs.	He et al. (2024); Maheshwari & Naik (2024)
Impact of digital literacy and collaboration	Digital literacy's role in facilitating business collaboration projects.	Abdullah et al. (2024); Viana & Gomes (2025)
	Does digital literacy improve preparedness for collaboration?	Akinwalere & Ivanov (2022); Blayone (2018)
Challenges and opportunities	Obstacles in learning and applying digital technologies.	Radzikhovska (2021); Rêgo et al. (2024)
	How digital literacy enhances competitiveness in digital transformation.	Farias-Gaytan et al. (2023); Ye et al. (2024)

3.4 Data Analysis

Using Braun and Clarke (2006) thematic analysis, the study examined how participants perceive and interpret digital literacy and collaboration experiences. The process involved reviewing recordings, transcripts, and chat logs to ensure accurate data categorisation. Thematic analysis followed Braun and Clarke (2006) six-phase approach: (1) data familiarisation; (2) initial coding; (3) theme identification; (4) theme review; (5) theme definition; and (6) report production. Coding was conducted manually by the lead researcher, followed by cross-review with a second researcher to ensure reliability. Any discrepancies were resolved through consensus, following best practices for analytical validation (Morrow, 2005). Triangulation was employed by cross-checking transcripts of focus groups with chat logs and field notes to confirm the consistency of emerging themes.

Initial coding identified key themes: (1) digital skills acquisition, (2) preparation for the digital labour market, (3) and academia-business collaboration (Table 3).

The themes were refined to align with participants' perspectives and research objectives. Manual coding was used instead of software, ensuring an in-depth understanding of participant experiences (Saldaña, 2025).

Zoom's automatic transcription feature assisted in categorising data for thematic analysis. Thematic saturation was reached when no new themes emerged from the participants' responses, a criterion supported by Palinkas et al. (2015), ensuring the adequacy of the sample and robustness of the findings.

Data were systematically organised into conceptual categories, supporting the research framework (Bryman, 2016).

Table 3. Categories and subcategories.

Category	Subcategory
Digital skills acquisition	Practical application of digital tools, technical competence in AI, IoT, and Big Data.
Preparation for the digital labour market	Perception of preparedness, exposure to real-world business scenarios.
Academia-business collaboration	Collaboration in internships and projects, partnerships for digital transformation efforts.
Challenges in digital literacy	Misalignment between curricula and market needs, limited access to digital tools in academia.
Opportunities in digital literacy	Emerging technologies in business enhanced employability through digital competence.

4 Results

4.1 Digital Skills Acquisition

Participants stressed the importance of practical digital skills for their future careers. Advanced tools such as data analytics and marketing automation were particularly valued in interactions with companies during internships and academic projects. One participant noted: "The digital skills we acquired throughout the course gave us a more practical view of what companies need in the digital market (…)".

However, all the students felt insufficient academic training and called for better curriculum alignment with market demands. Another participant stated: "Classes should focus more on acquiring and applying digital tools that solve concrete problems for companies (…)". This reflects existing research emphasising the need for continuous curricular adaptation to technological advancements (Abdullah et al., 2024; Rêgo et al., 2024).

4.2 Preparation for the Digital Labour Market

Students expressed concerns about whether their education had adequately prepared them for the digital job market. One participant remarked: "Although we have access to some digital tools during the course, I feel that we are not sufficiently prepared to work in an environment of rapid digital evolution". This aligns with research advocating for greater integration of digital skills into curricula to match technological changes (Calderon-Monge & Ribeiro-Soriano, 2023).

Internships and business projects were identified as essential in bridging the gap between theoretical knowledge and industry practice. Another participant noted: "I hope the internship will allow me to apply the digital skills I acquired in class and learn new ones". This supports findings that practical experience enhances students' confidence and employability (He et al., 2024).

Participants agreed that stronger links between academia and industry are essential for ensuring students are adequately equipped to contribute to companies undergoing digital transformation.

4.3 Academy-Business Collaboration

Most participants valued collaboration with companies, considering it instrumental in applying digital skills. One stated "(…) Projects, in some disciplines, with companies allowed me to understand how digital skills are used in the real world, something that would not be possible only with theoretical classes (…)".

However, participants noted inconsistencies in collaboration with limited opportunities or not aligned with real market needs. They highlighted the need to review and structure curriculum content to reflect industry demands. This perception aligns with research suggesting continuous academia-business collaboration is vital for digital transformation readiness (Petkovics, 2018; Radzikhovska, 2021).

4.4 Challenges in Digital Literacy

Students identified two major challenges: (1) limited access to up-to-date digital tools, leading to outdated knowledge and practices, and (2) the inability of certain modules to reflect the real-world needs of businesses. One participant remarked: "Some of the digital tools presented in our classes are already outdated compared to what companies are using (…)". Another challenge was the lack of practical opportunities, which could hinder their transition into the digital job market. Literature indicates that the gap between academic education and industry expectations remains a major barrier to effective digital transformation (Supriadi et al., 2024).

4.5 Opportunities in Digital Literacy

The adoption of emerging technologies such as AI, IoT, and Big Data analytics was seen to boost students' competitiveness. One participant stated: "I believe digital skills give us a competitive advantage, especially if we can apply them in real business scenarios (…)".

Participants also suggested that greater integration of practical projects and industry partnerships would better prepare students for digital market demands. Many agreed that, given their familiarity with technology, students could even assist companies in adopting new digital tools. This aligns with research suggesting that digital literacy enhances employability and fosters collaboration in digital transformation (Abdullah et al., 2024; Ye et al., 2024).

5 Discussion

The findings reaffirm the theoretical proposition (RP1) that digital literacy constitutes a critical enabler of students' preparedness for digital transformation. The results corroborate prior studies that emphasise the role of digital competencies in enhancing graduate

employability and alignment with market demands (Abdullah et al., 2024; Khan et al., 2022). In contrast with Blayone (2018), who argued that curricular reform alone is insufficient without systemic institutional change, the students' perceptions in this study suggest that even partial curricular integration of digital tools has perceptible effects on their market readiness. This nuance contributes to ongoing debates about the relative weight of curricular content versus institutional structure in preparing graduates for digital transformation.

Although the inclusion of digital competencies in academic programs has been acknowledged as an institutional priority, participants reported persistent gaps in their practical preparedness. This perception reinforces RP2 and aligns with findings by Supriadi et al. (2024) and Rêgo et al. (2024), who highlight the strategic role of structured academia-business collaboration in aligning curricula with real-world demands. Internships and collaborative projects emerged as key mechanisms for bridging these gaps, aligning with research highlighting experiential learning as fundamental for workforce readiness (Calderon-Monge & Ribeiro-Soriano, 2023; Petkovics, 2018). The findings also confirm the necessity of embedding digital competencies into curricula to prepare students for the evolving labour market (He et al., 2024; Ye et al., 2024). Participants recognised the strategic value of advanced digital skills, such as AI, IoT, and Big Data, in fostering innovation and supporting organisational adaptation to technological change.

Additionally, they emphasised the importance of structured academia-business partnerships in ensuring curricula aligning with organisational needs, allowing companies to contribute to curriculum development. These findings reinforce RP3, demonstrating that digital competencies when integrated with industry collaboration, enhance students' strategic and technical abilities while ensuring that curricula remain relevant to market expectations (Abdullah et al., 2024; Akinwalere & Ivanov, 2022; Radzikhovska, 2021; Szopik-Depczyńska et al., 2023; Ye et al.,2024). These findings suggest that integrating digital literacy into academic curricula should not be limited to technical proficiency but should also incorporate mechanisms that facilitate direct engagement with industry practices. Structured partnerships between academia and business and experiential learning opportunities contribute to closing the gap between education and labour market expectations. Ensuring that students engage with digital transformation processes throughout their training strengthens their ability to adapt to emerging industry requirements and contributes to organisational digital strategies.

6 Conclusion

6.1 Final Remarks

Students develop digital skills throughout their academic journey, yet they recognise gaps in technical proficiency and alignment with industry expectations.

The pace of technological evolution in organisations reinforces the need for structured academia-business collaboration to address these challenges. Incorporating AI, IoT, and Big Data into curricula enhances students' engagement with digital transformation processes, strengthening their readiness for the labour market and supporting organisations adapting to technological shifts.

However, technical knowledge alone is insufficient because practical learning opportunities are essential for bridging the gap between theory and application, ensuring that students can translate digital competencies into business contexts. Collaboration between academy and industry partners reinforces this connection and prepares graduates to contribute effectively to digital transformation.

In line with the four objectives that guided this study, the findings confirm that (1) students identify both challenges and opportunities in the acquisition of digital skills and involvement in academia-company collaboration; (2) their perceptions demonstrate a recognized gap between current curricula and the demands of the digital labor market; (3) evidence supports practical recommendations for curriculum adjustments, including updated digital tools and greater practice-business integration; and (4) the study contributes theoretically by positioning digital literacy as a key facilitator of cooperation between academia and business organizations, promoting digital transformation. These results reinforce the empirical and conceptual relevance of digital literacy in aligning academic training with the evolving requirements of digital markets.

6.2 Theorical and Practical Implications

This study highlights digital literacy as a central element in the intersection between education and digital transformation. The findings indicate that adapting academic curricula and strengthening academia-business engagement is essential for fostering technological innovation and organisational competitiveness.

From a practical perspective, embedding digital literacy into academic programs and fostering structured partnerships with industry ensures that students can develop technical expertise and gain practical experience relevant to market needs.

For academia, this may involve establishing annual curriculum reviews to integrate emerging digital trends and partnerships with local companies to ensure regular feedback on labour market needs. For companies, this may include structured internships and participation in co-designing of curriculum components that reflect current technological applications. Policymakers can also support joint programs through specific funding schemes, incentives for collaboration and investments in digital infrastructures that enable integration between academic and business systems.

Also, regular curriculum updates integrating technological advancements and experiential learning opportunities contribute to a more effective alignment with industry demands. In that way, the collaboration between academy and businesses can play a decisive role in shaping future professionals capable of leading digital transformation initiatives.

6.3 Limitations and Future Directions

This study focuses on final-year marketing students, excluding perspectives from business professionals and other stakeholders directly involved in digital transformation processes.

Future research should incorporate other qualitative and quantitative approaches to validate qualitative results and explore broader trends in digital skills development

and industry adoption of emerging technologies. Further studies could also assess how structured academia-business partnerships can be influenced by marketing orientation in digital transformation, ensuring that curricula remain dynamic and aligned with technological and market developments.

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